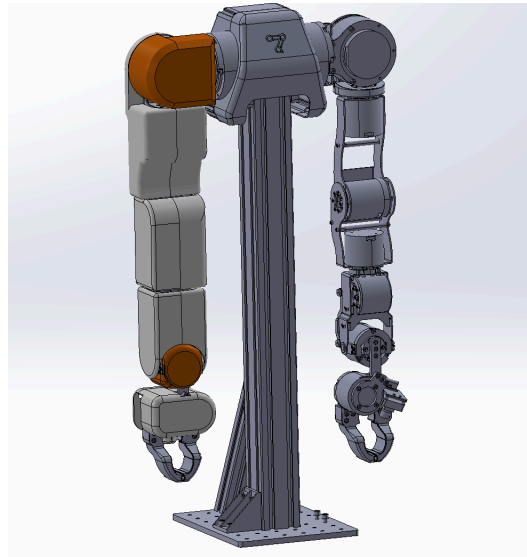


Designing a durable shell for the OpenArm 2.0

Kartik Ramachandran Design



Requirements/Assumptions

- Can not make any edits/changes to the open arm design itself
- I can remove the current basic shell in order to add mine
- Doesn't affect any joints or points of rotation
- Doesn't affect range of motion at any locations
- Account for wiring and storage for excess wiring at folding joints
 - Consider material to protect CAN wiring from possible EMI internally and externally due to CAN wiring's lack of protection
- Account for required heat displacing and access for repairs or damage

Understanding Joints

Joint 1 - Shoulder roll (base): full/near-full rotation about the vertical shoulder axis

Joint 2 - Shoulder pitch: rotation about the horizontal axis

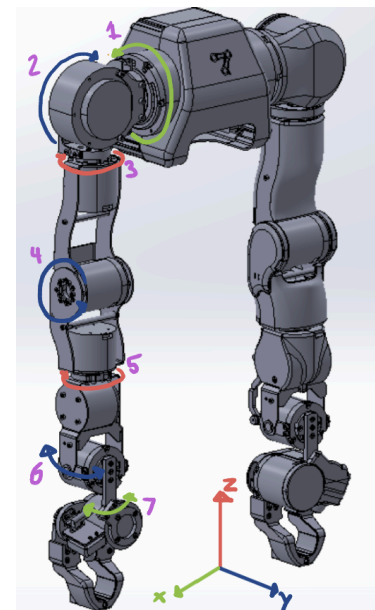
Joint 3 Shoulder yaw / upper-arm roll: rotation about the arm's long axis

Joint 4 - Elbow: single-axis bend

Joint 5 - Forearm roll: rotation about the forearm long axis

Joint 6 - Wrist pitch: rotation at the wrist

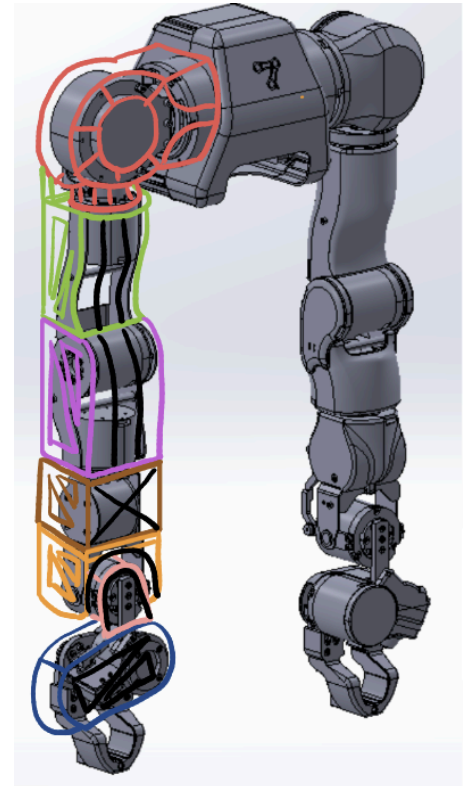
Joint 7 - Wrist roll: final roll about the tool axis



Concept Sketches

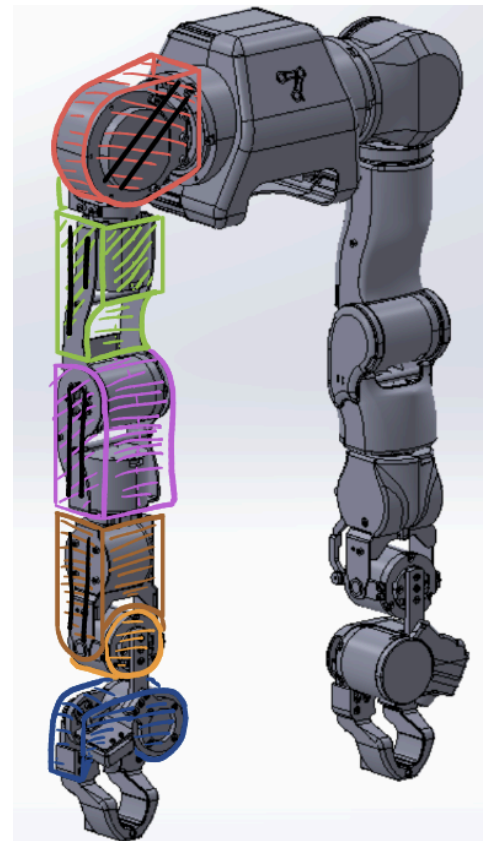
Manufacturing/Industrial Robot

- Aluminum Shell Primary
 - Made for durability and strength in different work environments
- Steel Support Rods to prevent bending in larger panels (Black Lines)
 - Worse COG but a required trade-off based on uses
- Copper wiring harness protection from powerful EMI sources
 - CAN wiring is susceptible to and EMI waves
- Larger more industrial style with designs for protection
 - Muscular view for visual satisfaction with in warehouses
 - Sharper chamfered corners for visual



Domestic Robot (3-D Model Designed)

- 3D-Printed Shell Primary (Injection Molding for Mass Production)
 - Light weight for improved COG and faster movements without tipping
- Larger panels supported by thin light weight aluminum rods
 - Black lines within sketch
- Aluminum wiring harness protection from household EMI (Microwave etc)
 - CAN wiring is susceptible to and EMI waves
- More friendly/softer design for internal home use
 - Soft filleted corners for visual appearance and safety within homes
 - Minimal gaps between multiple pieces for safety purposes

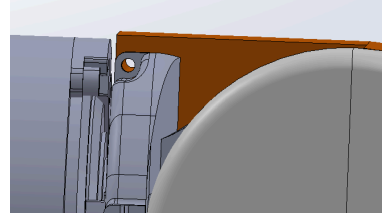


3D Model Design Decisions

Design Selection

OpenArm's own design videos and models have consistently shown the robot operating inside homes, so designing for domestic use reflects the most realistic scenario for a 7-DOF arm of this kind, a niche environment rather than a heavy industrial one.

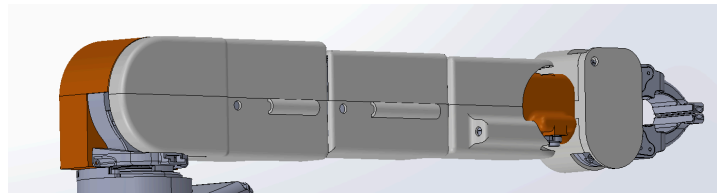
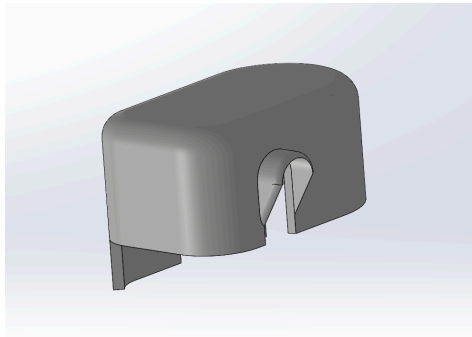
The shell follows that intent: 2.5–3 mm walls throughout, with deliberate clearance left between the shell and the internal components to allow airflow and heat dissipation. Panels mount using the arm's existing frame holes where available, avoiding any new features on the arm itself.



Camera and Wire integration

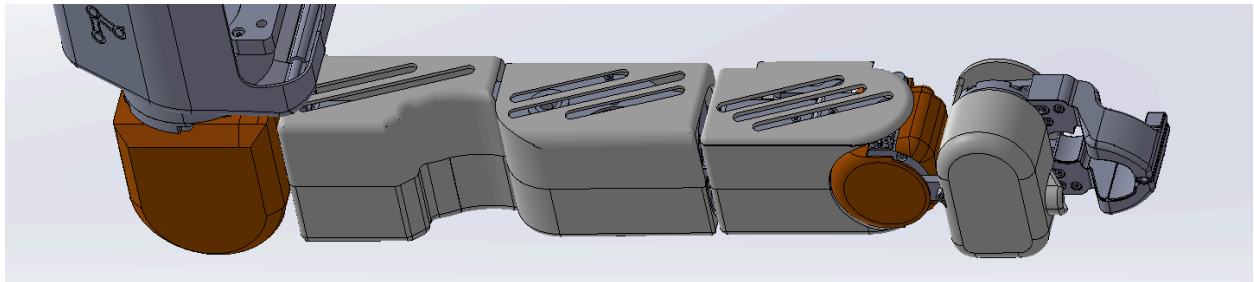
Both wrist cameras keep a full, unobstructed view of the gripper's contents, with the shell apertures shaped so they never cut into the field of view. The ceiling Arducam and the ZED stereo head remained unaffected by changes. The wiring runs down the back of the arm; because the back sits on the outer radius as the elbow and wrist flex, the cable path lengthens through those joints, so slack is left there to absorb the extra length without going into tension.

Dedicated gaps at each joint let the excess wire curl and sit passively rather than hang loose inside the shell. Since OpenArm uses CAN wiring, which is susceptible to EMI, the run is enclosed in a conductive metal harness (aluminum, copper, or steel) to shield it.



Paneling and Heat Rejection

The shell is split into two halves at every joint, which keeps assembly simple and lets any damaged panel be removed and replaced on its own. Each section also carries heat-rejection slots positioned near the motors to vent heat before it builds up inside the shell.



Manufacturing

For a single case basis, 3D printing would be suitable for the design. Since the design is just a shell, it is under no stress from any of the movements or structural pieces. The sole purpose of the shell is to protect what's located internally, and 3D print would be suitable for any possible damage to come from domestic uses.

Mobile Base

A four-wheel mecanum drivetrain is the best fit for this platform. A legged base would be unrealistic, the arm's length already raises the center of gravity, and legs would only add tipping risk. A four-wheel base is the simplest stable option and houses all the heavy electronics as low to the ground as possible, which is the main defense against pitching forward or back as the arm moves. Mecanum wheels give omnidirectional motion with no extra actuators, driving opposite wheels in opposing directions produces side-to-side travel and a motor-driven turntable at the base lets the whole upper body rotate 180°. I was only able to generate a concept sketch for this idea due to ongoing finals and lack of time.

The blue lines show a scissor lift that is supported by the purple bars and rotated at the red sprockets at the joints which would be chained doing so with a gear spinning below.

The base is a four wheel mecanum drive with all the electronics located as low to the ground as possible. The reason for this style of wheel frame compared to the conventional car is to prevent tipping forward and back when arms are moving.

